Offline
Calibration data
6 h
x(t)/x(6)
72 h
n units, m conditions
late label y ∈ {0, 1}
Posterior producer
p̂t ≈ Pr(y=1 | ℱt)
p*
calibration at the stopping time: κ(τ)
Rule generation
DP on 𝒟gen (Thm 1)
family 𝒢 = {πμ} of bands
fit-free: |p̂t − p*| ≥ m
Selection
fastest feasible rule π̂
α
π̂
risk + slack ≤ α on 𝒟sel; else the endpoint assay
Certification
independent units
Thm 3: w.p. ≥ 1 − δ
new conditions
Thm 4: in expectation, family fixed in advance
output: band (at, bt)
Online
same producer
new unit condition z
observe x(tj)
posterior p̂tj
p̂tj ∈ (atj, btj)?
no
stop at τ and decide δ = 1{p̂τ ≥ p*}
yes: wait one step, pay λ(tj+1 − tj)
deploy the certified band
stop
bt
at
p̂t
τ
time
Thm 2 excess realized cost over the best rule with the same information ≤ 2Lκ̄ + η, with κ̄ = supτ κ(τ) and η the surrogate slack
Cor 1 |believed − realized cost| ≤ L κ(τ) for every rule, so early-decision rules must be scored on realized cost